

Workshop Practice Lab:

Workshop Practice Lab is a vital part of the engineering curriculum that helps students develop practical skills and knowledge. It provides an opportunity for students to apply theoretical knowledge to real-world situations, fostering creativity and innovation. Here are some key points about Workshop Practice Lab:

- The Workshop Practice Lab is designed to equip students with the skills required to operate various hand and power tools. Students learn about the proper use of tools and equipment, safety measures, and maintenance procedures.
- The lab provides students with hands-on experience in designing and building projects, enabling them to understand the principles of engineering and mechanics. Students work on projects ranging from simple structures to advanced machinery, using a variety of materials such as wood, metal, and plastic.
- The Workshop Practice Lab is not only about building things, but it's also about problem-solving. Students learn to identify problems and develop solutions, working in teams to brainstorm ideas and develop prototypes.
- In the Workshop Practice Lab, students also learn about the importance of teamwork, communication, and leadership. These skills are essential in the workplace, and the lab provides an opportunity for students to develop them.
- Safety is a top priority in the Workshop Practice Lab. Students are required to follow safety procedures and wear protective gear at all times. The lab has safety protocols in place to prevent accidents and ensure that students are safe while working on projects.
- The Workshop Practice Lab is an integral part of the engineering curriculum, and it plays a significant role in the development of engineers. Students who participate in the lab gain valuable practical experience and are better prepared for the challenges of the workplace.

In conclusion, the Workshop Practice Lab is an essential component of the engineering curriculum that provides students with hands-on experience in designing and building projects. It helps students develop practical skills, problem-solving abilities, and teamwork skills. Safety is a top priority in the lab, and students are required to follow safety procedures at all times. The Workshop Practice Lab is an integral part of the engineering curriculum, and students who participate in the lab are better prepared for the challenges of the workplace.

S. No. Mechanical Workshop Duration

In mechanical engineering, workshops are essential for students to gain practical knowledge and hands-on experience. The duration of mechanical workshops can vary depending on the course and the level of study. Here are some points to keep in mind regarding the duration of mechanical workshops:

1. The duration of a mechanical workshop can range from a few weeks to several months, depending on the course requirements.
2. Typically, the duration of a mechanical workshop is determined by the number of practical sessions required for a particular course.
3. In some cases, the workshop may be conducted for a shorter duration if the course focuses more on theoretical concepts.
4. The duration of a mechanical workshop may also depend on the level of study. For example, a workshop for undergraduate students may be shorter than a workshop for graduate students.
5. The duration of a mechanical workshop may also be influenced by the availability of resources and facilities.
6. The workshop duration may also vary based on the type of workshop. For example, a workshop focused on welding may require a longer duration than a workshop focused on machining.
7. It's important to note that the duration of a mechanical workshop is not the only factor to consider. The quality of the workshop and the learning experience are equally important.
8. Students should make the most of the workshop duration by actively participating in practical sessions, asking questions, and seeking guidance from instructors.

In conclusion, the duration of a mechanical workshop can vary depending on several factors. However, regardless of the duration, it's crucial to make the most of the workshop and gain practical knowledge and experience.

1 Introduction to Mechanical Workshop Material, Tools, and Machines 3 Hrs

In this module, you will be introduced to the basic materials, tools, and machines used in a mechanical workshop. The following are the key points that you need to understand:

- The workshop is a place where mechanical engineers work to create and repair mechanical devices and machines.
- The workshop is equipped with various machines, tools, and materials required for the job.
- The workshop is divided into different sections, such as machining, welding, and sheet metal work.
- The machining section is equipped with machines like lathes, milling machines, drilling machines, and grinding machines.

- The welding section is equipped with machines like arc welding machines, gas welding machines, and TIG welding machines.
- The sheet metal work section is equipped with machines like shearing machines, bending machines, and punching machines.
- The workshop also contains various hand tools like hammers, screwdrivers, pliers, wrenches, and sockets.
- The materials used in the workshop include different types of metals, plastics, and composites.
- The workshop also has safety measures in place, such as safety goggles, gloves, and earplugs.
- It is important to follow the safety rules and regulations to avoid accidents and injuries.
- Proper maintenance of the machines and tools is also necessary to ensure their longevity and efficiency.
- In conclusion, the mechanical workshop is a place where engineers use machines, tools, and materials to create and repair mechanical devices and machines. Proper safety measures and maintenance are essential to ensure a safe and efficient working environment.

To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc) used in workshop.

As an aspiring engineer, it is important to have a thorough understanding of the layout and safety measures of a workshop. This knowledge not only ensures the smooth functioning of the workshop but also guarantees the safety of the workers. In addition to this, understanding the properties and uses of different engineering materials is crucial to the success of any project. Here are some key points to keep in mind when studying layout, safety measures, and different engineering materials used in the workshop:

Layout: - The layout of a workshop should be well-planned to ensure maximum efficiency and productivity. - The layout should also take into consideration the safety of the workers and ensure that hazardous materials and equipment are stored in designated areas. - The workshop should have clear markings and labels to indicate the location of emergency exits, fire extinguishers, and first aid kits.

Safety Measures: - It is important to always wear the appropriate personal protective equipment (PPE) while working in the workshop. - Ensure that all equipment is regularly inspected and maintained to prevent accidents. - Follow proper procedures for handling and disposing of hazardous materials.

- Familiarize yourself with the emergency procedures in case of accidents or injuries.

Engineering Materials: - Mild steel is a commonly used material due to its high strength, durability, and low cost. - Medium carbon steel is used for applications that require higher strength and hardness than mild steel. - High carbon steel is used for cutting tools and other applications that require high strength and wear resistance. - High-speed steel is used for cutting tools that require high heat resistance and wear resistance. - Cast iron is a popular choice for parts that require high compressive strength and wear resistance.

In conclusion, studying the layout, safety measures, and different engineering materials used in the workshop is essential for any aspiring engineer. By familiarizing yourself with these key concepts, you can ensure the successful completion of any project while prioritizing the safety of yourself and your fellow workers.

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section.

- Fitting tools: Fitting tools are used to hold and assemble various components. The commonly used fitting tools include spanners, pliers, screwdrivers, hammers, chisels, punches, and allen keys.
- Sheet metal tools: Sheet metal tools are used for the cutting, bending, and shaping of sheet metal. The commonly used sheet metal tools include snips, shears, nibblers, benders, rollers, and punches.
- Welding machines: Welding machines are used to join two or more metals by melting them together. The commonly used welding machines include arc welding machines, MIG welding machines, TIG welding machines, and spot welding machines.
- Cutting machines: Cutting machines are used to cut metals, plastics, and other materials. The commonly used cutting machines include plasma cutting machines, laser cutting machines, and water jet cutting machines.
- Measuring devices: Measuring devices are used to measure length, width, height, and other dimensions of the components. The commonly used measuring devices include tape measures, calipers, micrometers, and angle finders.
- Safety equipment: Safety equipment is used to protect the workers from hazards such as heat, sparks, fumes, and noise. The commonly used safety equipment includes welding helmets, gloves, aprons, safety glasses, and earplugs.
- Maintenance equipment: Maintenance equipment is used to maintain the tools and machines in good condition. The commonly used maintenance equipment includes lubricants, cleaning agents, and toolboxes.

- Power tools: Power tools are used to perform various tasks such as drilling, grinding, sanding, and polishing. The commonly used power tools include drills, grinders, sanders, and polishers.
- Hand tools: Hand tools are used to perform tasks that require precision and accuracy. The commonly used hand tools include files, rasps, saws, and planes.

In conclusion, the study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section is an essential aspect of the metalworking industry. By mastering the use of these tools and machines, workers can perform their tasks efficiently and safely.

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

In metrology, the least count of an instrument is the smallest value that can be measured accurately. The least count is essential to determine the precision of an instrument. In this experiment, we will determine the least count of three measuring instruments: Vernier calliper, vernier height gauge, and micrometer (screw gauge). We will also take different readings over given metallic pieces using these instruments.

Materials Required: - Vernier calliper - Vernier height gauge - Micrometer (Screw gauge) - Metallic pieces - Notebook and pen

Procedure:

1. Vernier Calliper:

- Take a metallic piece and place it between the jaws of the Vernier calliper.
- Close the jaws until they are in contact with the metallic piece.
- Take the reading of the main scale of the Vernier calliper.
- Take the reading of the Vernier scale of the Vernier calliper.
- Add the main scale reading to the Vernier scale reading to get the total reading.
- Repeat the process for at least three different metallic pieces and record the readings in the notebook.
- Calculate the least count of the Vernier calliper using the formula: Least count = (Smallest division of the main scale) / (Number of divisions on the Vernier scale)

2. Vernier Height Gauge:

- Place a metallic piece on the base of the Vernier height gauge.
- Bring the scribe in contact with the top of the metallic piece.
- Take the reading of the main scale of the Vernier height gauge.
- Take the reading of the Vernier scale of the Vernier height gauge.

- Add the main scale reading to the Vernier scale reading to get the total reading.
- Repeat the process for at least three different metallic pieces and record the readings in the notebook.
- Calculate the least count of the Vernier height gauge using the formula:
Least count = (Smallest division of the main scale) / (Number of divisions on the Vernier scale)

3. Micrometer (Screw Gauge):

- Place a metallic piece between the anvil and spindle of the micrometer.
- Turn the ratchet knob to apply a light pressure on the metallic piece.
- Take the reading of the main scale of the micrometer.
- Take the reading of the Vernier scale of the micrometer.
- Add the main scale reading to the Vernier scale reading to get the total reading.
- Repeat the process for at least three different metallic pieces and record the readings in the notebook.
- Calculate the least count of the micrometer using the formula: Least count = (Pitch of the screw) / (Number of divisions on the Vernier scale)

Observations:

Instrument	Metallic Piece 1	Metallic Piece 2	Metallic Piece 3
Vernier calliper	-	-	-
Vernier height gauge	-	-	-
Micrometer	-	-	-

Calculations:

- Least count of Vernier calliper = (Smallest division of the main scale) / (Number of divisions on the Vernier scale)
- Least count of Vernier height gauge = (Smallest division of the main scale) / (Number of divisions on the Vernier scale)
- Least count of Micrometer = (Pitch of the screw) / (Number of divisions on the Vernier scale)

Conclusion:

In this experiment, we determined the least count of three measuring instruments: Vernier calliper, vernier height gauge, and micrometer (screw gauge). We also took different readings over given metallic pieces using these instruments. The least count of each instrument was calculated using the formula. The least count of an instrument is essential to determine the precision of an instrument.

2 Machine Shop 3 Hrs

Machine shops are essential for manufacturing and producing various components and products. A machine shop is a facility where machines are used to shape, cut, and form materials to make products. In this section, we will discuss the basics of machine shops and how they operate.

1. Machine Shop Introduction
 - A machine shop is a facility where machines are used to shape, cut, and form materials to make products.
 - It is equipped with various tools and machines, including lathes, mills, grinders, and drill presses.
 - Machine shops can produce a wide range of products, from small components to large machinery.
2. Safety in Machine Shops
 - Safety is of utmost importance in a machine shop due to the nature of the work involved.
 - Workers must wear personal protective equipment, such as goggles, gloves, and respirators, to protect themselves from hazards.
 - The machine shop should have clear safety guidelines and procedures for workers to follow.
3. Common Machines in a Machine Shop
 - Lathes: A machine used to shape materials by rotating them against a cutting tool.
 - Mills: A machine used to cut and shape materials using rotating cutters.
 - Grinders: A machine used to smooth and shape materials using abrasive wheels.
 - Drill Presses: A machine used to drill holes in materials.
4. Machine Shop Processes
 - Machining: The process of shaping materials using cutting tools.
 - Turning: The process of rotating a material against a cutting tool to shape it.
 - Milling: The process of cutting and shaping materials using rotating cutters.
 - Grinding: The process of smoothing and shaping materials using abrasive wheels.
5. Machine Shop Materials
 - Metals: Common materials in machine shops, such as steel, aluminum, and copper.
 - Plastics: Used in various products due to their lightweight and durable properties.
 - Composites: Materials made from multiple components, such as fiberglass and carbon fiber.

In conclusion, machine shops are essential facilities for manufacturing and producing various components and products. Safety is of utmost importance, and

workers must follow clear guidelines and procedures. The common machines and processes in a machine shop include lathes, mills, grinders, drill presses, machining, turning, milling, and grinding. Materials used in machine shops include metals, plastics, and composites.

Demonstration of working, construction and accessories for Lathe machine Perform operations on Lathe - Facing, Plane Turning , step turning, taper turning, threading, knurling and parting.

In order to fully understand the working, construction and accessories of a Lathe machine, it is important to first understand the different operations that can be performed on a Lathe machine. Some of the key operations that can be performed on a Lathe machine include:

1. Facing: This is the process of cutting a flat surface on the end of a cylindrical workpiece. It is typically used to create a smooth finish on the end of a workpiece, or to ensure that the end of the workpiece is perpendicular to its axis.
2. Plane Turning: This is the process of cutting a cylindrical workpiece to a specific diameter. It is typically used to create a smooth finish on the outer surface of a workpiece, or to create a specific diameter for a given application.
3. Step Turning: This is the process of cutting a workpiece to a specific diameter at multiple points along its length. It is typically used to create a workpiece with a specific shape or to create a workpiece that will fit into a specific application.
4. Taper Turning: This is the process of cutting a workpiece to a specific diameter at one end and a different diameter at the other end. It is typically used to create a workpiece with a tapered shape, such as a cone or a frustum.
5. Threading: This is the process of cutting a series of helical grooves into the surface of a workpiece. It is typically used to create a threaded surface, such as a bolt or a screw.
6. Knurling: This is the process of cutting a series of small, raised ridges into the surface of a workpiece. It is typically used to create a surface with improved grip or to create a decorative pattern on the surface of a workpiece.
7. Parting: This is the process of cutting a workpiece into two or more separate pieces. It is typically used to create a workpiece with multiple parts or to create a workpiece with a specific shape.

In order to perform these operations on a Lathe machine, it is important to have a good understanding of the machine's construction and the various accessories

that can be used to perform these operations. Some key components of a Lathe machine include:

1. Headstock: This is the component of the Lathe machine that holds the workpiece and rotates it during operation.
2. Tailstock: This is the component of the Lathe machine that supports the other end of the workpiece and provides additional stability during operation.
3. Bed: This is the base of the Lathe machine and provides a stable platform for the other components.
4. Carriage: This is the component of the Lathe machine that moves along the bed and holds the cutting tool.
5. Cutting Tool: This is the tool that is used to perform the various operations on the workpiece.
6. Chuck: This is the component of the Lathe machine that holds the workpiece in place during operation.
7. Tool Post: This is the component of the Lathe machine that holds the cutting tool and allows it to be adjusted as needed.

By understanding the different operations that can be performed on a Lathe machine and the various components and accessories that are used to perform these operations, it is possible to fully appreciate the working, construction and capabilities of this important machine.

Fitting Shop

A fitting shop is a workshop where metal components are shaped, cut, and assembled using various tools and machines. It is an essential part of any manufacturing process, and its importance cannot be overstated. In this section, we will discuss the various aspects of a fitting shop.

Tools and Equipment

- Fitting shop requires an array of tools and equipment to carry out its operations effectively. Some of the essential tools include hammers, pliers, screwdrivers, wrenches, and vices. These tools come in various shapes and sizes, and each one has a specific purpose.
- The fitting shop also needs specialized machines such as drilling machines, lathe machines, milling machines, and grinding machines. These machines help in shaping, cutting, and finishing metal components to the required specifications.

Operations in a Fitting Shop

- The primary operations carried out in a fitting shop include marking, cutting, bending, drilling, and welding.
- Marking is the process of marking the metal components to indicate where cuts and bends need to be made.
- Cutting is the process of removing unwanted material from the metal component using saws or cutting machines.
- Bending is the process of bending the metal component to the required angle using a bending machine.
- Drilling is the process of making holes in the metal component using a drilling machine.
- Welding is the process of joining two or more metal components using heat and pressure.

Safety Precautions

- Safety is of utmost importance in a fitting shop, and workers need to take several precautions to avoid accidents.
- Workers should wear protective gear such as helmets, gloves, and goggles to protect themselves from flying debris and sparks.
- Machines should be regularly serviced and maintained to avoid accidents.
- Workers should be properly trained on how to use the tools and machines to avoid accidents.

Conclusion

In conclusion, fitting shop is an essential part of any manufacturing process, and it requires an array of tools, machines, and skilled workers to carry out its operations effectively. Safety precautions should be taken to avoid accidents, and workers should be properly trained to use the tools and machines.

1. Practice Marking Operations

In the field of computer programming, marking operations are essential for manipulating data and organizing it in a way that is easily accessible. Here are some key points to keep in mind when practicing marking operations:

1. Understand the purpose of marking operations: Marking operations help to manipulate data in a way that is efficient and effective for the programmer. They are used to organize data, perform calculations, and manipulate information in a variety of ways.

2. Familiarize yourself with the most commonly used marking operations: Some of the most common marking operations include arithmetic operations (addition, subtraction, multiplication, division), comparison operations (greater than, less than, equal to), and logical operations (AND, OR, NOT).
3. Practice using marking operations in different programming languages: Marking operations can be used in a variety of programming languages, including Java, Python, and C++. It's important to practice using these operations in different languages to develop a deeper understanding of their functionality.
4. Test your code thoroughly: Before deploying your code, make sure to test it thoroughly to ensure that the marking operations are working correctly. This can involve running test cases, debugging errors, and analyzing the output data.
5. Stay up-to-date with new developments in marking operations: As technology continues to evolve, new marking operations may be developed that can improve the efficiency and effectiveness of programming. It's important to stay up-to-date with these developments to ensure that your programming skills remain relevant and competitive.

By practicing marking operations regularly, you can develop a deeper understanding of how to manipulate data in a way that is efficient and effective for programming. This can improve the quality of your code and make you a more competitive programmer in the job market.

2. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.

When it comes to preparing a U or V-shape male-female work piece, it is essential to follow a series of steps to ensure that the final product is of high quality. Here are some of the steps involved in the preparation of a U or V-shape male-female work piece:

1. **Marking the workpiece:** The first step in preparing a U or V-shape male-female work piece is to mark the workpiece. This can be done using a scribe and a ruler. The marking should be done on both the male and female parts of the workpiece to ensure that they match up correctly.
2. **Sawing the workpiece:** Once the workpiece has been marked, the next step is to saw the workpiece. This can be done using a hacksaw or a power saw. It is important to ensure that the saw cuts are straight and accurate to ensure that the male and female parts of the workpiece match up correctly.
3. **Filing the workpiece:** After sawing the workpiece, the next step is to file the workpiece. This can be done using a hand file or a power file. The

filing should be done to achieve the correct U or V-shape for the male and female parts of the workpiece.

4. **Drilling the workpiece:** Once the filing is complete, the next step is to drill the workpiece. This can be done using a drill press or a hand-held drill. It is important to ensure that the holes are drilled accurately to ensure that the male and female parts of the workpiece match up correctly.
5. **Grinding the workpiece:** The final step in preparing a U or V-shape male-female work piece is to grind the workpiece. This can be done using a bench grinder or a hand-held grinder. The grinding should be done to achieve a smooth finish on both the male and female parts of the workpiece.

By following these steps, one can prepare a high-quality U or V-shape male-female work piece that is ready for use in a variety of applications. Remember to take your time and be accurate in each step to ensure a perfect final product.

4 Carpentry Shop 3 Hrs

The Carpentry Shop is a fundamental part of woodworking. In this shop, you will learn about various tools, equipment, and materials used in carpentry.

Here are some points to help you understand the Carpentry Shop:

- The Carpentry Shop is designed to assist you in creating woodworking projects. It is equipped with all the tools and machines that you'll need for your projects.
- You need to have a basic understanding of woodworking before you start working in the Carpentry Shop. You should know how to use a saw, a hammer, and other basic tools.
- Safety should always be your top priority when working in the Carpentry Shop. You should wear protective gear, such as goggles and gloves, to protect yourself from accidents.
- The Carpentry Shop has various tools and machines, such as saws, drills, sanders, and planers. You should know how to use each of these tools before you start working on your projects.
- You should also know about the different types of wood used in carpentry, such as softwood and hardwood. You can learn about the properties of different types of wood and their uses in carpentry.
- The Carpentry Shop has a variety of materials that you can use for your projects, such as nails, screws, glue, and wood filler. You should know which materials to use for different types of projects.
- In the Carpentry Shop, you can learn about different techniques used in woodworking, such as measuring and cutting wood, joining wood pieces, and finishing wood surfaces.

- You should also know about the different types of joints used in carpentry, such as butt joints, lap joints, and dovetail joints. You can learn about the strengths and weaknesses of each type of joint.
- Finally, you should practice your skills by working on various woodworking projects. You can start with simple projects, such as making a wooden box, and gradually move on to more complex projects.

In conclusion, the Carpentry Shop is an essential part of woodworking, and it provides you with the tools, equipment, and materials you need to create woodworking projects. By learning about the various tools, materials, and techniques used in carpentry, you can enhance your woodworking skills and create beautiful and functional projects.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint.

Carpentry is the art and craft of working with wood to create useful and decorative objects. It requires knowledge of tools, equipment, and different joints to create strong and aesthetically pleasing structures. In this study, we will delve into the world of carpentry tools, equipment, and different joints. We will also learn how to make three popular joints: Cross Half lap joint, Half lap Dovetail joint, and Mortise Tenon Joint.

Carpentry Tools and Equipment

1. Hand Tools

- Hammer
- Saw
- Chisels
- Screwdrivers
- Measuring tapes and rulers
- Planes
- Clamps
- Hand drill

2. Power Tools

- Circular saw
- Jigsaw
- Power drill
- Router
- Belt sander
- Table saw
- Miter saw

3. Safety Equipment

- Safety goggles
- Dust masks
- Ear plugs
- Work gloves

Carpentry Joints

1. Cross Half lap joint

- This joint is used to join two pieces of wood at right angles.
- It is made by cutting a slot in one piece of wood and a corresponding tab in the other piece of wood.
- The tab is then inserted into the slot and glued together.
- This joint is commonly used in making frames and boxes.

2. Half lap Dovetail joint

- This joint is commonly used in making drawers and boxes.
- It is made by cutting a dovetail shape on the end of one piece of wood and a corresponding dovetail shape on the other piece of wood.
- The two pieces are then fitted together to create a strong joint that resists pulling apart.

3. Mortise Tenon Joint

- This joint is used to join two pieces of wood at right angles.
- It is made by cutting a rectangular hole (mortise) in one piece of wood and a corresponding rectangular tab (tenon) on the other piece of wood.
- The tenon is then inserted into the mortise and glued together.
- This joint is commonly used in making furniture, doors, and windows.

In conclusion, carpentry is an important skill that requires knowledge of tools, equipment, and different joints. By learning how to make popular joints like Cross Half lap joint, Half lap Dovetail joint, and Mortise Tenon Joint, carpenters can create strong and aesthetically pleasing structures that stand the test of time.

5 Welding Shop 6 Hrs

Welding shop is an important aspect of the manufacturing and construction industry. It is essential that individuals who are interested in welding must learn the basics to ensure that they can carry out welding operations safely and efficiently. Here are some important points that one must keep in mind while learning about welding:

1. Safety precautions:

- Always wear protective gear like welding helmets, gloves, and jackets to prevent burns and injuries.
- Make sure the welding area is well-ventilated to avoid inhaling toxic fumes.

- Keep fire extinguishers and other safety equipment nearby.
 - Avoid standing in front of the welding arc to prevent eye damage.
2. Types of welding:
 - There are various types of welding processes like MIG, TIG, Stick, and Flux-cored welding.
 - Each welding process has its own advantages and disadvantages.
 - The choice of welding process depends on the type of material being welded, the thickness of the material, and the desired quality of the weld.
 3. Welding techniques:
 - To ensure a strong and quality weld, it is important to use the right welding technique.
 - Some commonly used welding techniques are butt joint, lap joint, and T-joint welding.
 - The choice of welding technique depends on the type of joint being welded and the thickness of the material.
 4. Material preparation:
 - Proper preparation of the material is essential for ensuring a strong and quality weld.
 - The material must be cleaned and free of rust, oil, and other contaminants.
 - The edges of the material must be beveled or chamfered to ensure proper penetration of the weld.
 5. Welding defects:
 - Welding defects like porosity, undercutting, and incomplete fusion can occur if the welding process is not carried out properly.
 - It is important to inspect the weld after completion to ensure that there are no defects.
 - Defective welds must be repaired before the material can be used.
 6. Welding symbols:
 - Welding symbols are used to indicate the type of weld and its location on a drawing.
 - It is important to understand welding symbols to ensure that the weld is carried out as per the design specifications.

By keeping these important points in mind, individuals can learn the basics of welding and carry out welding operations safely and efficiently.

Introduction to BI Standards and Reading of Welding Drawings. Practice of Making Following Operations: Butt Joint, Lap Joint, TIG Welding, MIG Welding

In this module, we will cover the basics of BI standards and reading welding drawings. We will also practice making several types of welding joints using TIG and MIG welding techniques. Here are the key points to keep in mind:

- BI standards: BI standards are a set of guidelines that ensure quality

control in welding. They cover everything from welder qualifications to weld inspection procedures. It is important to be familiar with these standards to ensure that your welding work meets industry standards.

- **Reading welding drawings:** Welding drawings are technical documents that provide instructions on how to weld a particular structure. They include information such as the type of weld, the size of the welding joint, and the welding process to be used. Being able to read these drawings is essential to ensure that you can accurately execute the welding work.
- **Butt joint:** A butt joint is a type of joint where two pieces of metal are joined together end-to-end. This joint is commonly used in structural welding and requires the use of filler material to complete the weld. Practice making butt joints using both TIG and MIG welding techniques.
- **Lap joint:** A lap joint is a type of joint where two pieces of metal overlap each other. This joint is commonly used in sheet metal work and requires careful alignment of the two pieces of metal to ensure a strong weld. Practice making lap joints using both TIG and MIG welding techniques.
- **TIG welding:** TIG welding, also known as Gas Tungsten Arc Welding (GTAW), is a welding process that uses a non-consumable tungsten electrode to produce the weld. This process is commonly used in industries such as aerospace and automotive due to its high precision and clean welds. Practice TIG welding on both butt and lap joints.
- **MIG welding:** MIG welding, also known as Gas Metal Arc Welding (GMAW), is a welding process that uses a consumable wire electrode to produce the weld. This process is commonly used in industries such as construction and manufacturing due to its high speed and efficiency. Practice MIG welding on both butt and lap joints.

With these key points in mind, you should have a solid foundation in BI standards, reading welding drawings, and making various types of welding joints using TIG and MIG welding techniques. Keep practicing and refining your skills to become a skilled welder.

6 Moulding and Casting Shop 6 Hrs

In this topic, we will discuss the moulding and casting process that is commonly used in manufacturing industries. The process involves creating a mould of a desired object and then pouring a liquid material into the mould to create a replica of the object.

Here are the key points to keep in mind:

- The moulding process involves creating a negative image of the desired object using a variety of materials such as silicone rubber, plaster, or clay.
- Once the mould is created, a casting material is poured into the mould to

create a replica of the original object. The casting material can be made of a variety of materials such as plastic, metal, or ceramic.

- The casting process is particularly useful in situations where the original object is difficult or expensive to produce using traditional manufacturing methods.
- The moulding and casting process can be used to create a wide range of objects such as toys, figurines, and even industrial parts.
- The process requires a certain level of skill and experience to ensure that the final product is of high quality and meets the desired specifications.
- Safety precautions must be taken when working with casting materials, as they can be toxic and can cause health problems if not handled properly.
- It is important to choose the right casting material for the job to ensure that the final product meets the desired specifications and is durable enough to withstand normal use.

Overall, the moulding and casting process can be a very useful tool in the manufacturing industry, allowing for the creation of complex objects that would be difficult or expensive to produce using traditional manufacturing methods. However, it is important to approach the process with caution and to take the necessary safety precautions to ensure that the final product is of high quality and meets the desired specifications.

Introduction to Patterns:

- Patterns are the models or replicas of the final product that is to be casted.
- They are made from materials such as wood, metal or plastic.
- They are used to create a mould for the casting process.

Pattern Allowances:

- Pattern allowances are the extra dimensions added to the pattern to compensate for the shrinkage that occurs during the casting process.
- They are different for each material used in the casting process.

Ingredients of Moulding Sand:

- Moulding sand is used to create a mould for the casting process.
- The ingredients of moulding sand include silica sand, clay, water and additives.
- The type and amount of additives used depend on the type of casting process being used.

Melting Furnaces:

- Melting furnaces are used to melt the metal or material that is to be casted.
- They are available in different sizes, depending on the size of the castings.
- The temperature and time required for melting depends on the type of material being used.

Foundry Tools and Their Purposes:

- Foundry tools are used for different stages of the casting process.
- Some of the commonly used tools include crucibles, ladles, tongs, sprue cutters, moulding boxes, and sand rammers.
- These tools are used for tasks such as melting, pouring, and shaping the mould.

Demo of Mould Preparation and Aluminum Casting Practice:

- This practice involves preparing a mould for the casting process using a pattern.
- The mould is filled with molten aluminum to create the final product.
- The process involves using foundry tools and techniques.

Study and Preparation of Mould for Plastic:

- This practice involves preparing a mould for the casting process using plastic as the material.
- The process involves creating a pattern, preparing the moulding sand, and pouring the plastic into the mould.
- The process requires knowledge of pattern making, moulding sand preparation, and casting techniques.

7 CNC Shop 6 Hrs

- CNC stands for Computer Numerical Control, which is a manufacturing process that involves using computer software to control machinery, such as lathes, mills, routers, and grinders.
- The 7 CNC Shop 6 Hrs program is a training course that teaches individuals how to operate and program CNC machines.
- The program is designed to provide students with the skills and knowledge needed to work in the manufacturing industry, specifically in shops that use CNC machines.
- The course is six hours long and covers a variety of topics, including:
 - Understanding the basics of CNC machines
 - Reading and interpreting blueprints and schematics
 - Programming CNC machines using G-code
 - Troubleshooting common problems with CNC machines
- The program also includes hands-on training, where students will have the opportunity to work with real CNC machines in a shop environment.

- Upon completion of the program, students will be able to:
 - Set up and operate CNC machines
 - Write and edit CNC programs using G-code
 - Identify and troubleshoot common problems with CNC machines
- The 7 CNC Shop 6 Hrs program is an excellent option for individuals who are interested in pursuing a career in manufacturing, as CNC machines are widely used in this industry.
- The course is also ideal for individuals who are looking to expand their skills and knowledge in the field, as well as for employers who want to provide their employees with additional training.
- Overall, the 7 CNC Shop 6 Hrs program is a comprehensive and valuable training course that provides students with the skills and knowledge needed to succeed in the manufacturing industry.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines.

- A CNC machine, or Computer Numerical Control machine, is a manufacturing tool that is controlled by a computer program.
- CNC machines are used to perform a variety of operations on metal components, such as cutting, drilling, milling, and turning.
- The main features of a CNC machine include a computer control unit, a cutting tool, a workpiece holder, and a spindle.
- The computer control unit is responsible for controlling the movements of the cutting tool and workpiece holder.
- The cutting tool is used to remove material from the workpiece, while the workpiece holder holds the workpiece in place during machining.
- The spindle is responsible for rotating the cutting tool at high speeds to perform the machining operation.
- CNC machines can be equipped with a variety of accessories to enhance their capabilities, such as tool changers, rotary tables, and probing systems.
- Tool changers allow the machine to automatically switch between different cutting tools during a machining operation.
- Rotary tables allow the workpiece to be rotated during machining, which is useful for creating complex geometries.
- Probing systems are used to measure the position and orientation of the workpiece, which is useful for ensuring accurate machining.
- CNC machines can be programmed using a variety of programming languages, such as G-code and M-code.
- G-code is a standardized programming language that is used to control the movements of the cutting tool and workpiece holder.
- M-code is used to control auxiliary functions of the machine, such as coolant flow and spindle speed.
- To perform different operations on metal components using a CNC ma-

chine, the operator must first create a computer program that specifies the desired machining operations.

- The program is then loaded into the CNC machine, and the machine is set up with the appropriate cutting tools and accessories.
- Once everything is set up, the machine can be operated to perform the desired machining operations on the metal component.

8 To prepare a product using 3D printing 3 Hrs

Preparing a product using 3D printing is a fascinating process that requires specific skills and knowledge. The following points will guide you on how to prepare a product using 3D printing in 3 hours:

1. Determine the product design: The first step is to create or obtain a 3D model of the product design. This can be done using 3D modeling software or by downloading a pre-made design from online repositories.
2. Prepare the 3D printer: Ensure that the 3D printer is in good working condition and has the required materials such as filament or resin. Also, make sure that the print bed is level and clean.
3. Slice the 3D model: Slicing is the process of dividing the 3D model into layers that the 3D printer can understand. This can be done using slicing software, and the settings should be adjusted to match the 3D printer's specifications.
4. Load the 3D model: After slicing, load the 3D model into the 3D printer's software and adjust the print settings such as layer height, infill density, and print speed.
5. Start the print: Once the 3D model is loaded, start the 3D printing process. This can take several hours, depending on the complexity of the design and the speed of the 3D printer.
6. Post-processing: After the 3D printing is complete, remove the product from the print bed and clean it up. This may involve removing support structures, sanding, and painting the product.
7. Quality check: Finally, inspect the product for any defects or inconsistencies. If any are found, make the necessary adjustments and reprint the product.

By following these steps, you can prepare a product using 3D printing in just 3 hours. However, it is essential to note that 3D printing requires patience and attention to detail to produce high-quality products.

Total 33 Hrs

Here are the key topics that you should study for a total of 33 hours:

1. **Calculus** (10 hours)

- Limits and continuity
- Differentiation and integration
- Applications of derivatives and integrals
- Sequences and series

2. **Linear Algebra** (8 hours)

- Linear equations and matrices
- Vector spaces
- Linear transformations
- Eigenvalues and eigenvectors

3. **Probability and Statistics** (8 hours)

- Probability theory
- Random variables
- Descriptive statistics
- Inferential statistics

4. **Discrete Mathematics** (7 hours)

- Sets and functions
- Logic and proofs
- Combinatorics
- Graph theory

5. **Numerical Analysis** (5 hours)

- Interpolation and approximation
- Numerical integration and differentiation
- Root-finding algorithms
- Matrix computations

By studying these topics for a total of 33 hours, you will have a solid foundation in mathematics that will serve you well in various fields such as engineering, finance, and data science. It is important to practice problems and exercises to reinforce your understanding of the concepts. Good luck with your studies!

Course Outcome:

The course is designed to provide students with a deep understanding of the subject matter and equip them with the necessary skills to excel in their chosen field. By the end of the course, students should be able to:

- Demonstrate a comprehensive understanding of the fundamental concepts and principles of the course.
- Apply critical thinking and problem-solving skills to analyze and interpret complex information related to the course.
- Develop effective communication skills to present their findings and ideas both orally and in writing.

- Evaluate and synthesize information from various sources to create well-reasoned arguments and solutions.
- Work collaboratively with others to achieve common goals and objectives.
- Apply ethical and professional standards in their work and interactions with others.
- Utilize technology and other relevant tools to enhance their learning and performance in the field.
- Continuously engage in self-reflection and self-improvement to enhance their knowledge and skills in the field.

Overall, the course outcome is to prepare students for success in their chosen field by providing them with a solid foundation of knowledge, skills, and values. With dedication and hard work, students can achieve their professional goals and make meaningful contributions to their respective fields.

The Students Will Be Able To

- Use various engineering materials, tools, machines, and measuring equipment at a Blooms Level CO1.
 - Understand the properties and characteristics of different engineering materials such as metals, polymers, and composites.
 - Identify and select the appropriate tools and machines for a given manufacturing process.
 - Accurately measure and record dimensions using measuring equipment such as vernier calipers, micrometers, and depth gauges.
- Perform machine operations in lathe and CNC machine at a Blooms Level CO2.
 - Understand the basic principles of lathe and CNC machines.
 - Set up and operate a lathe machine to perform turning, drilling, and threading operations.
 - Set up and operate a CNC machine to perform milling, drilling, and cutting operations.
- Perform manufacturing operations on components in fitting and carpentry shop at a Blooms Level CO3.
 - Understand the basic principles of fitting and carpentry shop operations.
 - Identify and select the appropriate hand tools and power tools for a given manufacturing process.
 - Accurately measure and mark dimensions on the workpiece.
 - Perform operations such as cutting, filing, drilling, and sawing to manufacture components.
- Perform operations in welding, molding, casting, and gas cutting at a Blooms Level CO4.
 - Understand the basic principles of welding, molding, casting, and gas cutting.
 - Identify and select the appropriate welding technique based on the

- type of material and joint.
- Understand the molding and casting process and perform operations to manufacture components.
- Set up and operate a gas cutting machine to perform cutting operations on metal sheets and plates.
- Fabricate a job by 3D printing manufacturing technique at a Blooms Level CO5.
 - Understand the basic principles of 3D printing technology.
 - Identify and select the appropriate 3D printing process based on the type of material and complexity of the job.
 - Design a 3D model of the job using computer-aided design (CAD) software.
 - Set up and operate a 3D printer to fabricate the job according to the design.

Reference Books:

Here are some reference books that you can use to supplement your studies:

- “The Elements of Style” by William Strunk Jr. and E.B. White: This book is a classic guide to writing well, covering everything from grammar to style.
- “The Chicago Manual of Style”: This is a comprehensive guide to writing and publishing, with detailed advice on everything from grammar to citations.
- “The Oxford English Dictionary”: This is the definitive dictionary of the English language, with over 600,000 words and phrases, as well as historical and etymological information.
- “The Merck Manual”: This is a comprehensive guide to medical conditions and their treatments, used by healthcare professionals around the world.
- “Roget’s Thesaurus”: This is a classic thesaurus, giving you synonyms and antonyms for every word in the English language.
- “The Cambridge Handbook of Psychology and Economic Behaviour”: This is a comprehensive guide to the intersection of psychology and economics, covering topics such as decision-making, risk-taking, and social influence.
- “The Handbook of Mathematical Functions”: This is a reference book for mathematicians, covering everything from basic algebra to complex analysis.
- “The Handbook of Chemistry and Physics”: This is a comprehensive guide to the physical sciences, covering everything from atomic and molecular structure to thermodynamics and quantum mechanics.

- “The Handbook of Social Psychology”: This is a comprehensive guide to the field of social psychology, covering topics such as social cognition, interpersonal relationships, and group dynamics.

These reference books can be a valuable resource for anyone looking to deepen their understanding of a subject or improve their skills in a particular area.

1. Workshop Practice, H S Bawa, McGraw Hill

Workshop practice is a fundamental aspect of learning mechanical engineering. It involves the application of theoretical principles in the practical setting of a workshop. H S Bawa’s book, Workshop Practice, published by McGraw Hill, provides an in-depth guide to workshop practices. Here are some of the essential points covered in the book:

- **Safety Precautions:** Safety is the top priority in any workshop. The book covers various safety precautions that need to be taken while working with different types of equipment and tools. It also emphasizes the importance of wearing protective gear such as gloves, goggles, and helmets.
- **Hand Tools:** Hand tools are essential in any workshop. The book covers the different types of hand tools and their uses. It also includes the proper handling and maintenance of hand tools.
- **Machine Tools:** Machine tools are an essential part of any workshop. The book covers the different types of machine tools, their uses, and their maintenance. It also includes the proper use of machine tools and the safety precautions that need to be taken while using them.
- **Measuring Instruments:** Accurate measurements are crucial in workshop practice. The book covers the different types of measuring instruments and their uses. It also includes the proper handling and maintenance of measuring instruments.
- **Welding and Joining:** Welding and joining are essential skills in workshop practice. The book covers the different types of welding and joining techniques and their applications. It also includes the safety precautions that need to be taken while welding and joining.
- **Sheet Metal Work:** Sheet metal work is an important aspect of workshop practice. The book covers the different types of sheet metal work and their applications. It also includes the proper handling and maintenance of sheet metal tools.

In conclusion, Workshop Practice by H S Bawa, published by McGraw Hill, is an essential guide for anyone interested in learning workshop practices. It covers all the essential topics, including safety precautions, hand tools, machine tools, measuring instruments, welding and joining, and sheet metal work. The book provides a comprehensive understanding of workshop practice and is an excellent resource for students and professionals alike.

Mechanical Workshop Practice, K C John, PHI

This book covers a wide range of topics related to mechanical workshop practice. Here are some of the key points to keep in mind:

- Safety should always be the top priority in any workshop. Make sure to follow all safety guidelines and wear appropriate personal protective equipment (PPE) when working with machinery and tools.
- It is important to understand the different types of hand tools and their uses. Some common hand tools include wrenches, pliers, hammers, and screwdrivers.
- Power tools, such as drills and saws, can be dangerous if not used properly. Always read the manufacturer's instructions before operating any power tool.
- Proper maintenance of equipment is essential for safe and efficient operation. This includes regular cleaning, lubrication, and inspection of machinery.
- Understanding measurement and calibration is also important in mechanical workshop practice. Make sure to use the correct measuring tools, such as micrometers and calipers, and know how to read and interpret measurements.
- In addition to hands-on skills, it is important to have knowledge of materials and their properties. This includes understanding the differences between various metals, plastics, and other materials commonly used in mechanical workshop practice.
- Finally, good communication and teamwork are essential in any workshop environment. Make sure to communicate clearly with colleagues and work together to achieve common goals.

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary, Media promoters and Publications

Workshop Practice Vol 1 and Vol 2 are two books written by HazraChoudhary, published by Media Promoters and Publications. These books are a great resource for anyone interested in learning about workshop practices. Here are some key points to keep in mind:

- Workshop Practice Vol 1 covers the basics of workshop practices, including tools, equipment, and safety measures. The book provides detailed information on various tools used in a workshop, such as chisels, hammers, and wrenches. It also covers the different types of machines used in a workshop, such as lathes, milling machines, and drilling machines.
- Workshop Practice Vol 2 covers more advanced topics in workshop practices, such as welding, forging, and casting. The book provides detailed

information on different types of welding processes, such as arc welding, MIG welding, and TIG welding. It also covers the basics of forging and casting, including the types of materials used and the different techniques involved.

- Both books are well-illustrated with diagrams and photographs, making it easy for readers to understand the concepts being discussed.
- Workshop Practice Vol 1 and Vol 2 are suitable for both beginners and advanced learners. The books are written in a simple and easy-to-understand language, making it easy for readers to grasp the concepts being discussed.
- The books are an excellent resource for students studying mechanical engineering, as well as for professionals working in the field. The books cover a wide range of topics related to workshop practices, making them a comprehensive guide for anyone interested in the subject.
- Workshop Practice Vol 1 and Vol 2 have been widely acclaimed by readers and experts in the field. The books are highly recommended by professionals in the industry, and have been used as a reference by many students and researchers.

In conclusion, Workshop Practice Vol 1 and Vol 2 are two excellent resources for anyone interested in learning about workshop practices. The books cover a wide range of topics, from basic tools and equipment to advanced welding and casting techniques. The books are well-illustrated, easy to understand, and suitable for both beginners and advanced learners.

4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication.

CNC (Computer Numerical Control) is a highly advanced technology used in manufacturing processes. It involves the use of computer systems to control machine tools, such as lathes, mills, routers, and grinders. A CNC machine is programmed to perform specific tasks with a high degree of accuracy and precision. The following points provide an overview of CNC fundamentals and programming:

- CNC machines use a programming language called G-code to control their movements. G-code is a standardized language used in the industry, and it consists of a series of commands that tell the machine what to do.
- The G-code program is created using a CAM (Computer-Aided Manufacturing) software. The program includes information about the toolpath, cutting speed, feed rate, and other parameters that the machine will follow.
- The CNC machine operator loads the G-code program into the machine's memory and initiates the machining process. The machine executes the

program, moving the tool along the programmed path and cutting the workpiece according to the specified dimensions.

- CNC machines can perform a wide range of operations, including drilling, milling, turning, and grinding.
- CNC programming requires a good understanding of mathematics, geometry, and trigonometry. The programmer must be able to calculate angles, distances, and other parameters accurately.
- The programmer must also have a thorough understanding of the machine tool and its capabilities. They need to know how to select and set up the cutting tools, how to adjust the cutting speed and feed rate, and how to use coolant or lubricant to improve the cutting performance.
- CNC machines are widely used in the manufacturing industry because they offer several advantages over traditional manual machines. They are faster, more accurate, and can produce complex parts with high precision and consistency.
- Learning CNC programming requires practice and patience. It is essential to start with simple programs and gradually build up to more complex ones. There are many resources available online and in books that can help beginners learn CNC programming.

In summary, CNC fundamentals and programming are essential skills for modern manufacturing. CNC machines offer high accuracy, speed, and consistency, making them ideal for producing complex parts with tight tolerances. Learning CNC programming requires a good understanding of mathematics, geometry, and the machine tool's capabilities. With practice and dedication, anyone can learn CNC programming and become a skilled CNC machinist.